

MOHAMED SALAMA

Senior Software Engineer | Tech Lead | Distributed Systems & FinTech

@ mohamedsalama.dev@gmail.com +32 472 12 61 28 Brussels, Belgium
mohamed.salama mohamed.salama kaggle.com/mohamed.salama



CORE COMPETENCIES

Leadership & Management

- Technical Leadership (12+ devs)
- Engineering Team Lead
- Technical Hiring
- Intern Mentoring
- Agile / SCRUM
- Code Review Standards
- Runbook Authoring
- Cross-Functional Collaboration

System Architecture

- Distributed Systems
- Microservices
- Event-Driven Architecture
- High Availability Design
- Scalability Patterns
- Circuit Breakers
- API Gateway
- Service Mesh

Backend Engineering

- Node.js (NestJS / Express)
- Java (OOP, Design Patterns)
- Python, TypeScript
- REST / GraphQL APIs
- gRPC / Protobuf
- WebSockets
- RabbitMQ, Kafka

Cloud & DevOps

- AWS (EC2, ECS, RDS, S3, Lambda)
- GCP (GKE, Compute Engine)
- Kubernetes, Docker
- Terraform (Production)
- CI/CD (GitLab, GitHub Actions)
- Nginx

PROFESSIONAL SUMMARY

Senior Software Engineer and Tech Lead with **7+ years** architecting mission-critical systems across FinTech, high-frequency trading, and automotive verticals. Led distributed engineering teams of **12+ developers**, conducted technical hiring, and mentored interns while delivering measurable system performance and reliability improvements. Currently, Architecting and optimizing full-spectrum Banking Payment Infrastructure – Covering Retail, corporate, and Regulatory Payment Operations – in a regulated EU financial institution. Expert in Node.js, Java, gRPC, Kubernetes, and distributed systems. Brussels-based with full EU work authorization. Multilingual: English, French, Arabic, German. Entrepreneurial background in digital product development.

EXPERIENCE

Senior Software Engineer — Payment Systems

KEYTRADE BANK

October 2024 – Present Brussels, Belgium

- Led payment systems engineering for a regulated EU financial institution, architecting full-spectrum retail and corporate banking payment infrastructure including Standing Orders, Instant Payments, Memo Payments, SWIFT transfers, and SEPA Direct Debits.
- Delivered **50% performance improvement** by re-engineering standing order execution from legacy SQL-based jobs to an event-driven cronjob architecture with exponential backoff retry logic, reducing system load and enabling horizontal scaling.
- Designed and implemented independently a circuit-breaker “skip-record” mechanism that improved payment pipeline success rate from **92% to 99%**, preventing single transaction failures from cascading and blocking entire batch processing queues.
- Built observability platform replacing basic database logging with self-triggered success metrics and real-time anomaly detection using **Dynatrace** and **Elasticsearch**, significantly reducing payment failure detection and response time.
- Reduced new engineer onboarding time by **30%** through establishment of runbooks, architecture documentation, and structured mentoring protocols.
- Applied Infrastructure as Code using **Terraform** in production for cloud infrastructure management on AWS.

Node.js React.js RabbitMQ Kafka gRPC Protobuf AWS
PostgreSQL Redis Terraform Dynatrace Elasticsearch

Software Engineer — HFT Analytics Platform

FEATURES ANALYTICS

July 2023 – September 2024 Nivelles, Belgium

- Designed full-stack trading analytics platform bridging high-throughput computational backend with intuitive Vue.js dashboards, delivering actionable market intelligence to institutional clients.
- Optimised AI model analysis pipeline using decision tree algorithms, enabling near real-time processing of trade data for high-frequency analytics workloads.

Java (OOP) Python Node.js Vue.js PostgreSQL AWS

Observability & SRE

- Dynatrace, Elasticsearch
- AWS CloudWatch, Datadog
- ELK Stack
- Distributed Tracing
- Incident Management
- Performance Optimization

Data & Storage

- PostgreSQL, MySQL, MSSQL
- MongoDB, ArangoDB
- Redis
- SQL Optimization
- Query Performance Tuning
- Caching Strategies

FinTech & Banking

- Payment Systems Architecture
- Standing Orders, Instant Payments
- SWIFT, SEPA, Direct Debits
- Banking Compliance
- Payment Gateway Integration
- Regulatory Financial Systems

CERTIFICATIONS



AWS Certified Cloud Practitioner

Issued July 2023 – Valid through July 2026

Currently advancing toward next AWS certification tier.

LANGUAGES

Arabic
English
French
German



Arabic: Native | English: Fluent | French: Conversational | German: Basic

Full-Stack Software Engineer & Team Lead

SYLNDR

September 2022 – July 2023

Cairo, Egypt

VC-backed (\$12.6M pre-seed) automotive FinTech marketplace — one of Africa’s largest pre-seed rounds — processing \$20–25K average vehicle transactions, achieving ~10× sales growth during tenure.

- Led distributed engineering team of **12+ developers** across backend, frontend, and mobile, establishing code review standards, CI/CD pipelines (GitLab), and agile sprint rituals.
- Architected secure, highly available multi-cloud microservices platform (AWS + GCP) integrating **15+ third-party APIs** including Paymob (payments), Cequens/Twilio (SMS communications), and a custom scheduling system for vehicle inspection appointments.
- Deployed production Kubernetes clusters with automated CI/CD release pipelines, eliminating manual deployment processes and enabling a consistent release cadence.
- Designed and built custom vehicle inspection scheduling and calendar system, coordinating warehouse visits and home delivery logistics for buyers.
- Conducted technical hiring interviews and mentored interns, contributing to team growth during a period of approximately tenfold sales growth.

Node.js

NestJS

React.js

React Native

PostgreSQL

Redis

MongoDB

Kubernetes

AWS

GCP

GitLab CI

Full-Stack Software Engineer

EPICORE

April 2021 – September 2022

Cairo, Egypt

- Built 3 full-stack e-commerce applications with monolithic backend architecture, delivering end-to-end features in an agile environment.
- Designed and implemented scalable REST and GraphQL APIs applying industry best practices for API design, query optimisation, and database performance.
- Applied performance tuning techniques across PostgreSQL and MongoDB to improve response times and data throughput.

Node.js

Express

GraphQL

PostgreSQL

MongoDB

Full-Stack Developer

LOCUSKEY

June 2019 – March 2020

Cairo, Egypt

- Architected backend server infrastructure integrating payment gateways, push notification services, and third-party booking APIs for an early-stage tourism platform.
- Designed and managed deployment pipeline, ensuring reliable release processes for platform services.
- Mentored junior developers on backend architecture, API design, and deployment best practices.

Node.js

PostgreSQL

Vue.js

AWS (EC2, RDS)

ENTREPRENEURIAL VENTURES

Travel Guide & Planner

Digital travel planning product.

E-Commerce Platform

Early-stage experimental e-commerce platform.

EDUCATION

Bachelors of Computer Science & Engineering

German University in Cairo

📅 September 2016 – July 2021

📍 Cairo, Egypt

R&D Engineer (Bachelor Thesis)

DHBW Stuttgart

📅 March 2020 – August 2020

📍 Stuttgart, Germany

- Developed an intelligent athlete performance tracking system using Computer Vision and Machine Learning to optimize cyclist training behavior.
- **Technologies:** Python, OpenCV, NumPy, Pandas, Keras

PERSONAL PROJECTS

Rental Management App

TypeScript

A versatile full-stack property rental management application for landlords and property owners.

MovieTrailerEngine

Vue.js / Node.js

Distributed movie discovery platform handling concurrent users with microservices architecture.

Apartment Search Engine

JavaScript

Property search and filtering platform.